Operating Systems LAB (LAB-217)
Credit Hrs: **LAB 1 (Pre-Req : Data Structures)**
Lab 4 : **Hiding, UnHiding, Zipping, Unzipping Files**
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Section : **C**
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Objectives:

The objective of this lab session is to enable students to hide and unhide files. Students will learn about the importance of this feature in a linux environment and they'll be elaborated about some important hidden files in our directory system. Students will also learn how to zip, unzip files and directories.

Program Learning Outcomes (PLOs):

- PLO1: Academic Education
- PLO2: Knowledge for Solving Computing Problems
- PLO3: Problem Analysis
- PLO4: Design/Development of Solutions
- **PLO5: Modern Tool Usage**
- PLO6: Individual and Team Work
- PLO7: Communication
- PLO8: Computing Professionalism and Society
- PLO9: Ethics
- PLO10: Life-long Learning

CLO–PLO Mapping with BT Levels

CLO	Learning Outcome	BT Level	Mapped PLOs
CLO1	**Practice** Linux commands and utilities to gain proficiency with the command-line interface and use it effectively for system management tasks, basic scripting, software installation, and other essential operations in a Linux environment.	A5	PLO5

Practice Exercises

Before we begin today's graded lab exercise:

- Open your **terminal** as we start the session.
- Follow along and replicate everything I do and try to internalize the commands and concepts.
- This will prepare you for today's **graded exercise**.

We practiced following commands in previous weeks ;

`echo, whoami, hostname, pwd, cd, ls, clear, script, touch, tee, mkdir`

`which, whatis, man, help, wc, cp, mv, head, tail,  more, less, grep`

and explored their powers by using a lot of cool switches along with them and piping together multiple commands to perform multiple things as a single unit to do a lot of useful activities.

These are the commands for this week ;

`zip , unzip, mv, nano, pico, ls -a, history`

and we shall explore the use of different switches along with these commands for modifying their default behaviours and adding more functionality to them.

We will continue playing around with files and learn about hiding, unhiding and displaying hidden files in our directories. We will also learn to zip, unzip files in a linux environment from the terminal.

Your lab task for today is primarily based upon these topics so make sure to practice with me so you can easily attempt your lab task.

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Graded Exercises

Task 1:

- **This time submission will be in the form of screenshots (not script sessions)**

- Display a short introduction message containing your name and roll number on the terminal.

- Open your terminal and navigate in the **User's home directory** and display the contents of this directory such that the **hidden files** of this directory also display on your terminal.

- Create a directory named after your own name and within that directory create a new text file. Display the contents of this directory for confirmation that this file is created and present here.

- Use Nano Text Editor to write some secret text in this file and save and exit it.

  (You might consider roasting the Noob GUI Users in that text.)

- Now display the content of this file to show that the text is successfully added to it.

- Now **hide** this file. Then display the contents of your working directory to verify that this file is hidden.

- Now **unhide** this file and again display the contents of your directory to verify that it's not a hidden file anymore.

- Display the contents of this file to show that the data contained in the file is preserved after hiding and unhiding it.

- Again **hide** this file. Then display the contents of your working directory to verify that this file is hidden. **The GUI User shouldn't see this.**

Task 2

- Display a short introduction message containing your name and roll number on the terminal.

- Navigate to the directory named after you, remove everything from it so you have a clean environment for your working.

- Create 10 **text files** starting with the name Section-C and move them into a new directory named after Section-C. You might need to create a new directory with this name.

- Create 10 **text files** starting with the name Section-A and move them into a new directory named after Section-A. You might need to create a new directory with this name.

- Create 10 **text files** starting with the name Section-B and move them into a new directory named after Section-B. You might need to create a new directory with this name.

- Now zip these 3 directories into 3 zip-files named **Section-C.zip, Section-A.zip,Section-B.zip** such that these files are moved into these zipped files and no text file should remain in your current working directory.

- Display the contents of your current working directory now, it should only contain the zipped files and no other file.

- Now extract these zipped files in a common new directory named **All-Sections-Extracted** which should be created within your current working directory.

- Now display the contents of your current directory and it should only contain these zipped files and **All-Students-Extracted** directory.

- Remove this zip file from your current directory and navigate into the directory where all extracted files are created.

- Display the contents of this directory to verify that we have all text files extracted here successfully.

Submission Guidelines

- Take screenshots of your work and submit a pdf document of those screenshots. You can use any online tool for creating pdf from these screenshots.

- Screenshot of 1st task should be clearly separated from the 2nd in your pdf document.